Bergeon Group: Screen Processing & Tracking Pipeline

Project Overview

A proposed automated pipeline and web/mobile application designed to streamline the lifecycle of printing screens. The system replaces manual tracking with a barcode scanning system to manage screen drop-offs, processing, and invoicing accurately.

Proposed System Workflow & Features

- **Service Intake:** Log incoming screens and assign them for either Cleaning or Burning..
- **Logistics & Routing:** Track the initial pickup step (e.g., from the Receiving Room) and maintain an automated manifest of what needs to be returned.
- **Barcode Tracking System:** Scan screens to instantly pull up required specifications, including Screen Name, Size, and Associated Artwork.
- **Production Logging:** Track internal production metrics such as:
    - Turnaround deadlines (e.g., 1-day SLAs).
    - Technician accountability ("Who burned it?").
    - Material tracking ("What emulsion?").
    - Drop-off/Pickup personnel and timestamps.
- **Automated Invoicing:** Automatically calculate final costs based on the initial request, generating an invoice that details the exact number of screens burned and returned.

UI/UX Design & Architecture Approach

The application can be built using a robust API backend (FastAPI) and a relational database (PostgreSQL/Neon) to handle the relational tracking of screens, invoices, and users. The frontend will feature a clean, minimal interface utilizing a Black, Cream, and Gold color palette, styled with sharp, solid borders (no box shadows) for high visibility and ease of use in a production environment.

#	Topic	Discovery Question	Clarification Target
1	**Hardware & Scanning**	Will the team use standard smartphones (camera scanning) for the Barcode System, or do you have dedicated hardware	Device compatibility & barcode printing specs

#	Topic	Discovery Question	Clarification Target
		scanners? Do we need to build a feature to generate and print these barcodes?	
2	**Artwork Integration**	For tracking "Artwork," do we need the application to store and display actual image files, or just reference a file name/number?	Asset storage & display requirements
3	**Material Specs**	Should we pre-populate dropdown menus for specific Emulsions used during the burn process, or keep it as an open text field?	Data input constraints & validation